

Epic 003: TeXlyre Citation Assistant

Status: In Progress **Created:** 2026-04-11

Goal

Get the `./texlyre` fork running as Kuhn's near-term editor shell, then add the first real editor-native research assistant flow: a `/cite` slash command that suggests grounded citations for the claim in the previous sentence. The initial design should assume a small browser-resident model for low-latency language tasks, while citation lookup and verification rely on authoritative, user-configurable scholarly sources to minimize hallucinated references.

Context

We already have a local TeXlyre fork in the repo, and it gives us several useful starting points:

- A working browser-based LaTeX/Typst editor with collaboration and preview
- Existing bibliography integrations, including OpenAlex and Zotero
- An embedded Draw.io viewer that can later support `/diagram`
- A plugin-oriented architecture that appears suitable for editor-side extension work

The orchestration layer is still undecided, so this epic should avoid coupling the first assistant workflow to a heavy backend agent runtime. The first milestone is narrower: prove that Kuhn can invoke an editor command, inspect nearby writing context, search configured scholarly sources, rank candidate papers conservatively, and help the user insert a real citation without fabricating metadata.

For on-device intelligence, we want to start by evaluating Qwen 3 1.7B in-browser for lightweight intent parsing, context reading, query generation, and optional result summarization. However, the model must not be treated as a source of truth for bibliography data. `/cite` should only present references that were actually retrieved from configured sources and passed metadata validation.

Key Decision Factors

- **Grounding rigor** — `/cite` must never invent papers, authors, DOIs, PMIDs, or venues
- **Hint handling** — user-provided clues should improve retrieval without being treated as facts
- **Editor integration** — command flow must feel native inside TeXlyre rather than bolted on

- **Local responsiveness** — simple command understanding should work quickly in-browser
- **Configurable sources** — users should be able to enable, disable, and prioritize citation providers
- **Source quality** — default providers should cover scientific and technical literature well
- **Bibliography insertion** — accepted citations must map cleanly into `.bib` workflows and editor autocomplete
- **Privacy model** — keep local context local when possible and make external lookups explicit
- **Extensibility** — the same foundation should later support `/diagram` and `/search`

Initial Scope

- Bring up the TeXlyre fork as a working Kuhn development baseline
- Identify extension points for slash-command-style editor actions
- Evaluate an in-browser Qwen 3 1.7B runtime for lightweight assistant tasks
- Design and implement `/cite` first
- Support optional user hints in the command input, such as likely authors, years, venues, or keywords
- Define a provider abstraction for authoritative literature sources
- Start with a configurable default provider set:
 - PubMed
 - arXiv
 - medRxiv
 - bioRxiv
 - PsyArXiv
 - IEEE Xplore
 - OpenAlex

Non-Goals

- Finalize the full multi-agent orchestration architecture
- Ship `/diagram` or `/search` beyond design and interface scaffolding
- Solve all citation styles and bibliography-management edge cases in the first pass
- Trust the LLM to generate bibliographic records without external verification

Acceptance Criteria

- ☐ `texlyre` runs locally in this repo with Kuhn-specific setup notes documented

- ☐ Editor extension points for slash commands and citation insertion are mapped
- ☐ In-browser Qwen 3 1.7B feasibility is evaluated with concrete performance and packaging constraints
- ☐ `/cite` flow is specified end-to-end, from sentence capture to validated reference insertion
- ☐ `/cite` supports optional user hints that influence query generation without bypassing validation
- ☐ Citation providers are configurable and at least the initial default set is supported at the interface level
- ☐ Suggested references are grounded in retrieved source metadata rather than model-only generation
- ☐ Evaluation plan exists for citation quality, hallucination resistance, and UX usefulness

Stories

#	Story	Status	Size
001	Bootstrap the TeXlyre fork for Kuhn development	done	M
002	Map editor extension points for slash commands and citation insertion	done	M
003	Spike in-browser Qwen 3 1.7B integration	done	L
004	Design the research assistant command UX	done	M
005	Define configurable citation providers and validation rules	done	L
006	Implement grounded retrieval for <code>/cite</code>	done	XL
007	Integrate <code>/cite</code> into the editor and bibliography workflow	ready	XL

#	Story	Status	Size
008	Evaluate citation quality and hallucination resistance	draft	M
009	Fix CORS failures for arXiv and PsyArXiv providers	ready	M
010	Improve citation search quality with NLP and multi-variant queries	draft	L